

May 4, 2026

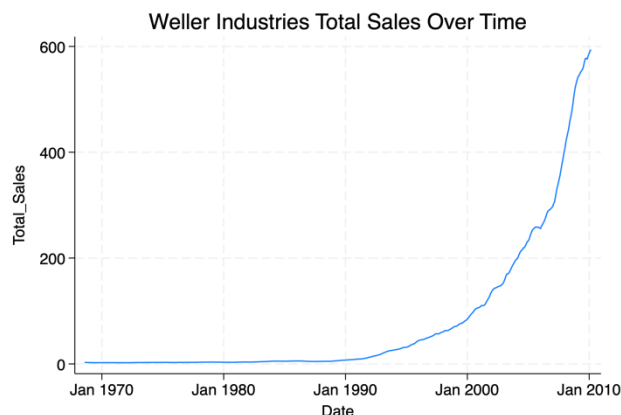
Weller Industries Sales Forecast Report

This report provides model estimation and forecast predictions for sales figures from Weller Industries based on 500 observations of monthly data between 1968 and 2010. There are five total sections in the report, with a sixth additional section for conclusive thoughts.

1. Data Transformation

To begin with, I destrunged the data and transformed the 'Date' variable into a proper date format. I then generated a new variable called 'date_monthly', which will be the time variable for analysis; a time variable is necessary in time series analysis so that the software you use is able to register how far apart observations are across time; This is especially helpful for data with missing values, so that way data is not treated as if it occurred over one continuous length of time. This is not an issue with the data you provided me, however, as this data is continuous monthly observations from August of 1968 to March of 2010. Once the data was in the proper format for Stata usage, I tested its suitability for time series analysis in its raw state. In order to predict an effective predictive model, the data must appear to look random, with no obvious trends or seasonality. Figure 1 depicts the raw data you provided me, plotted over time.

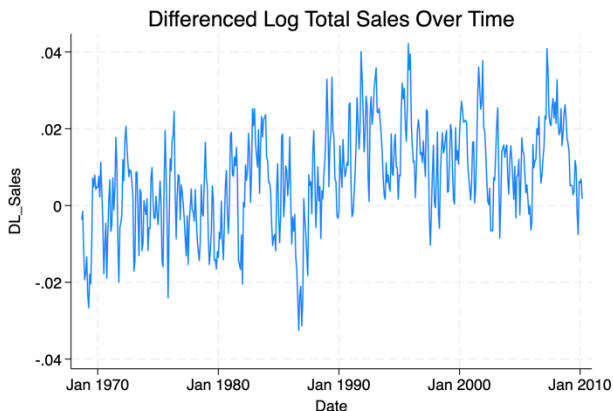
Figure 1: Raw Total Sales Data (Nominal Terms)



The raw data follows a strong exponential trend, therefore making it unsuitable for time series analysis in this state. I began by turning the nominal sales data into real sales data, which will be more useful once forecasted. Then, to address the upward trend, I took the log of all observations. After taking the log, the data unfortunately still displayed a strong upward trend. The next transformation I tried was differencing, which takes the current value of an observation and subtracts the value of the observation one period prior (in this case,

since the data is monthly, it would subtract the value from the previous month). This helped make the data appear more random, however it did not help fix the trend issue, as the sharp increase still remained after differencing. In this format, the data does not pass one of the key assumptions we need to run even a simple linear regression, because as time goes on, the variance (how much the data fluctuates), grows larger, which is not what we want to see. To fix this issue, I took the difference of the logged data to see how this changes the visualization. This transition was successful in making the data appear in a proper time series format, as shown below in Figure 2. This is the final transformation of the data, which now visually appears suitable for time series analysis. The variance and mean both appear roughly constant, and the data is mostly varied around zero which is a great sign. One element that I was cautious of at this point in the transformation stage was the slight increase around January of 1990; I worried that this may affect how the data would perform in a stationarity test, which I will now discuss in the model estimation stage.

Figure 2: Transformed Real Sales Data, Logged and Differenced



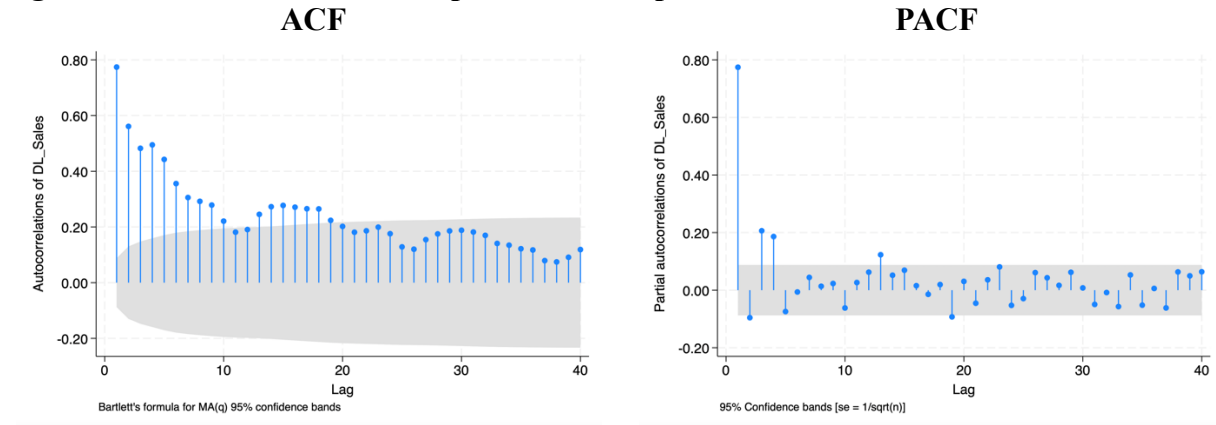
2. Estimation and Testing for a Set of Possible Models

Once the data was in the proper format, I evaluated and tested the most appropriate model(s) with which to forecast future sales values. I set aside the last 100 observations of sales data to use as a test sample once I had estimated a model. A crucial step before estimating is to determine if data is stationary. In time series analysis, stationarity ensures that data is reliable and won't yield spurious

regression, such as a false correlation between two completely unrelated variables. In order to be stationary at a weak but valid baseline, data must have a constant mean, constant variance, and have autocovariance dependent only on the number of lags in the model and not time. Lags refer to how many periods of frequency (in this case months) we are delaying data values by, so that we may estimate a relationship between a variable and itself from previous periods. This way, we can use that relationship to predict values in future lags, also known as forecasting. Stationarity, therefore, ensures that relationships estimated from the past will hold relevance in the future. By logging and differencing the data, I attempted to make it appear more stationary, but I also ensured it was statistically stationary as well. To check for stationarity, I first ran an Augmented Dickey Fuller test with 12 lags, because the data is monthly. Augmented Dickey Fuller tests check if data has a unit root, which is present in non-stationary data and means that data follows a completely random, unpredictable path. The p-value for the ADF test was 0.01, meaning we can reject at 95% confidence that there is a unit root in this data. I next ran a Phillips-Perron test, which also checks for a unit root, as an additional level of robustness on these results. This test also provided sufficient evidence that there is no unit root in the data. However, just because the data does not have a unit root does not automatically imply it is stationary, so I therefore ran another test to support the results from the ADF and Phillips-Perron tests. The final test I ran to check stationarity was a KPSS test, with a maximum of 12 lags. The test statistics for all 12 lags were beyond the critical value of 0.739 to reject at the 99% threshold; however, unlike the ADF and Phillips-Perron tests, this rejection is not supportive of the data being stationary, but rather non-stationary. This result is confusing because the data visually appears stationary and both unit root tests cleared any evidence of a unit root, but I am not counting the rejection from the KPSS test as a marker for non-stationarity. This is because occasionally when data has too much autocorrelation—meaning it is highly correlated with past values of itself—the KPSS test can mistake it for non-stationarity. This is because the KPSS test assumes that data is weakly dependent, while this data appears to be quite strongly dependent. To test this theory, I ran a correlation function on total sale and all lags confirmed high autocorrelation. Though proceeding with caution is highly recommended when estimating models after these results, I believe that the conclusions from the transformed data visualization, ADF and Perron tests, KPSS and correlation test all point to this data as stationary but highly autocorrelated. Once I was almost completely certain of stationarity, I proceeded with model estimation. For the first stage of

estimating a model for the data, I created plots for the autocorrelation and partial autocorrelation functions (ACF and PACF). These plots display how time series data is related to its values from past periods, and can help determine what model should be used to estimate.

Figures 3 and 4: ACF and PACF plots from sample data



The slow decline in autocorrelation in the ACF as lag increases indicates a trend in this data, which matches the plot of the raw data in Figure 1; further, the fact that the coefficients are all positive indicates the autocorrelation is positive, which the raw data corroborates. With the gradual decline of autocorrelation coefficients in an ACF plot and the sharp cutoff displayed in the PACF plot, these suggest this data would be best suited for estimation with a subsection of ARMA(p,q) model, where p and q denote the number of lags. In this stage of estimation, I was inclined to believe that this data would be estimated best with an AR(p) model instead of ARMA because of the ACF plot; the ACF of an ARMA model will typically decay towards 0 with possible oscillation after initial mixed behavior, while the ACF of an AR model will not display any initial mixed behavior, and I believed this ACF plot therefore matched an AR description better. Regardless, I tested multiple ARMA models for the potential case that AR is not the better fit. To test model efficiency, I fitted multiple different potential models and estimated their AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion), which are values used to determine and evaluate how well different models comparatively fit the same set of data. The model with the lowest AIC and BIC is considered the best fit for the data; however, if one model has the lowest AIC while another has the lowest BIC, both remain potential candidates for model selection and additional information is necessary to make a decision. I ran all possible ARMA model combinations using 0-3 for lags. The full list of models is available in the accompanied do. file of this report. The model with the lowest AIC was ARMA(3,3), while the model with the lowest BIC was ARMA(1,3). Based on these criteria, I proceeded with estimation as a decision between these two models.

After selecting two finalist models, I tested their residuals to determine if they contained white noise or not. White noise refers to error terms from a model that have an approximate mean of 0 and display no autocorrelation. If the residuals from these models are white noise, then there is no more variability left to capture in the model, and all relevant data patterns have been extracted and synthesized. I ran a Ljung-Box test to check for white noise, which I also ran in the initial stages of data transformation when I tested the autocorrelation for sales. I chose to run it

again on both of the models to check individual efficiency. Both models failed to reject the null hypothesis, meaning their residuals are white noise.

In addition to total sales data for Weller Industries, I was also provided data on CPI, Interest Rates, and Production Rate. The inclusion or exclusion of these variables is crucial to estimation, because their relationship with total sales numbers can completely alter the type of model one should use. To check for stationarity, I transformed these variables in the same way I did to total sales (CPI and interest rate can coexist as nominal and real as long as both are in a stationary format). I ultimately chose to omit all three variables from the final model. After plotting all three across time with the transformed sales data, I determined that production and sales were highly visually correlated, which matches economic intuition. If I included production data into the final model, it would incur too much multicollinearity and I would need to change models. However, because the primary goal of this report is to forecast sales outcomes, I chose to omit production from the final model. I also chose to omit CPI because it was used to transform Sales from nominal to real values, so I once again worried about incurring multicollinearity if I included it. Lastly, I chose to omit interest rates because they are not likely to cause total sales changes because the product has one price. I will elaborate more on interest when I discuss the Lucas Critique.

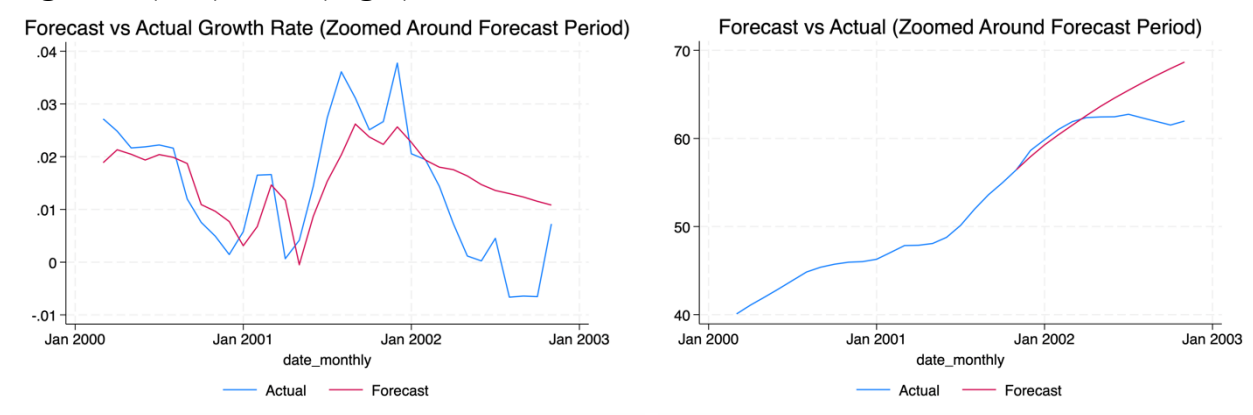
3 Out-of-Sample RMSE results

Now that I had estimated two models, I wanted to decide which would be a better fit for this data. I tested the Mean Squared Errors (MSE) for both ARMA models and created ACF plots for the residuals. Both models had relatively low mean squared error, which is excellent, because it means their errors did not deviate that far from 0. However, the second model, ARMA(3,3), appeared as a more preferable model for two reasons. First, it had a slightly smaller MSE than the ARMA(1,3) model (0.0000626 versus 0.0000637); secondly, it failed to reject the residual white noise test at a greater rate than the other model (p-values of 0.19 and 0.07), implying the residuals for this model are more statistically aligned with white noise than the other. For these reasons, I chose to forecast the data using the ARMA(3,3) model.

4 Model Summary and Proposed Forecast

Once I had chosen a model, I predicted 100 future values and compared them to the actual data from that time period. The plot, available in the accompanying .do file, shows the forecasted data appears to become a very flat line next to the actual values across an increasing time horizon. While this may be concerning, it should be noted that this is actually a normal response for forecasted time series data. This is because this model is forecasting growth rates (Figure 5), not just levels or individual values. The reason the forecast asymptotically approaches zero is because ARMA models approach zero as the time horizon increases, which additionally matches the assumptions for stationary data. Because the model estimates log values that were differenced, forecasts will converge to the unconditional mean of the data, which in this case is zero. To show total sales value forecasts, I un-differenced and exponentiated the data, of which is plotted in figure 6. Since the model uses 3 lags and I used 12 lags for multiple tests to account for this being monthly data, it is important to keep in mind that this forecast should be used to understand predictions for up to 12 months (one year) in the future. After these plots, I created 12 brand new predicted observations to forecast a full year ahead using all 500 observations.

Figures 5 (Left) and 6 (Right): Forecasted Growth Rates and Sales Next to Actual Data



5 Relevance of the Lucas Critique

Lastly, I would like to briefly discuss the relevance of the Lucas Critique—which involves the instability of historical predictions in response to policy changes—to this data. While I think the Lucas Critique should always be evaluated and considered when forecasting time series data, I do not think it applies to these sales figures. The primary reason for this is I forecasted sales rates based on past sales values, which has nothing to do with policy changes. In general, ARMA models are not used to model policy shocks and are instead designed to predict data using past values. For robustness, I ran and plotted the correlation between the sample sales data and interest rate data I was provided, and the correlation was essentially non-existent. This adds on to my previous reasoning for why I chose to omit interest rates from my final model: I didn't believe they were relevant enough to be included. So to recap, I don't believe the Lucas Critique should be considered because this predictive model involves analyzing past statistical patterns, not structural behavior in response to other economic changes.

6 Conclusion

To summarize, this report develops and estimates an ARMA(3,3) model to forecast Weller Industries total sales figures across a time horizon of one year using monthly sales data. Interest rate and production, while helpful for other estimations, were selectively excluded from the model because of their irrelevance and collinearity with sales, respectively. CPI was used to transform sales data from nominal to real. I chose an ARMA(3,3) model and believe it is a more robust prediction choice than the 'rule of thumb' because it accounts for past values of sales fluctuation to predict where figures are likely to go. The 'rule-of-thumb' Weller Industries has been using can be dangerous because it leaves no room to predict negative values. For example, as Figure 5 shows, the sales growth rate decreased dramatically after January 2002, and while the ARMA model did not predict as sharp of a decrease, it correctly predicted that the growth rate would decrease, directly contradicting the rule of thumb approach. That approach will ultimately make sales values and rate much more difficult to predict because it ignores the potential for negative outcomes which should be taken into account to develop the most accurate model. I believe the ARMA(3,3) model I estimated is the best current predictor for future sales outcomes because it passed every test for suitability to fit the data, and it has the smallest Mean Squared Error, meaning did not leave any variability to be predicted after it fit the data.